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ORIGINAL RESEARCH

Beneficial Effects of Multi-Micronutrient Supplementation with Collagen Peptides on Global Wrinkles, Skin Elasticity and Appearance in Healthy Female Subjects

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ABSTRACT

Introduction: With ageing, collagen production slows down, leading to wrinkle appearance and loss of elasticity. Replenishing key structural molecules through oral supplementation is a promising strategy that complements the topical delivery of cosmetic products and creates a holistic skincare regimen. The present study assessed the effectiveness of a food supplement with collagen peptides, vitamins and minerals in improving the quality of the skin and general wellbeing of healthy women.

Methods: This was an open-label study of 135 women aged between 45 and 65 years. A 3-month treatment phase followed a 4-week washout phase, with visits scheduled at baseline and after each month of treatment. Subjects received three tablets of Richelet Skin Renewal daily. The primary outcome was change from baseline to month 3 in global wrinkles score by expert grader analysis. Secondary outcomes included changes in skin elasticity and other skin attributes, product assessment via self-perception questionnaires and total antioxidant status.

Results: A total of 116 subjects completed the study. The mean global wrinkles score indicated a statistically significant decrease from 5.9 at baseline to 5.0 at month 3 ($p<0.0001$), with 83.6% of subjects showing an improvement; significant changes were reported at all intermediate visits. The increase in skin elasticity was also statistically significant (R2 score 0.74 at month 3; $p<0.0001$). All subjects (100%) demonstrated significant improvements in skin texture, skin tone evenness, skin radiance and overall skin quality at the month 3 visit.

Conclusions: The study product achieved statistically significant, noticeable effects on global wrinkles, skin elasticity and a range of skin attributes after 3 months of use in healthy women. These results strengthen the evidence for supplementation of collagen peptides and other micronutrients as an effective component of anti-ageing skincare.

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Key Summary Points

Skin ageing is a multi-factorial process marked by a decrease in the production of key structural molecules such as collagen and hyaluronic acid. Hallmarks of skin ageing include wrinkle formation, loss of elasticity and loss of hydration.

Daily skincare regimens can have a significant long-term impact on the overall skin quality. In contrast to traditional topical skincare products, dietary supplements are able to target the deepest skin layers from the inside.

The present study investigated the effectiveness of oral supplementation with collagen peptides, hyaluronic acid and micronutrients in improving skin quality and delaying the visible signs of ageing. The limitations of the study design were the lack of a control group and no control of nutrition (although use of other supplements was forbidden).

Statistically significant, noticeable effects on global wrinkles, fine lines, skin elasticity, skin firmness, skin tone evenness, skin texture, radiance and overall skin quality after 3 months of supplementation in healthy female subjects were recorded by an expert grader. These benefits were also noticeable and meaningful to participants, as confirmed by subjective data from self-perception questionnaires.

These results strengthen the evidence for supplementation of collagen peptides, hyaluronic acid and other micronutrients as an effective approach for improving skin outcomes in a clinical setting, with the potential to benefit everyday practice and beauty regimens on a broader scale.

INTRODUCTION

Skin Ageing and Renewal

The skin is the largest organ of the human body, with a surface area of 1.5–2 m^2 [1]; it is composed of three layers: epidermis, dermis and hypodermis, each with well-defined roles [1–3]. Collagen is the main structural protein in the dermis (75% of total dry weight [4]) and is responsible for most of its mechanically supportive function [5, 6]. Another critical component of the dermis is elastin, which provides resilience and elasticity to the skin [4, 5, 7].

Human skin is particularly sensitive to ageing, a complex process that involves a combination of extrinsic and intrinsic factors [8–10]. Constant exposure to ultraviolet (UV) sunlight, environmental pollutants, endocrine disruptors and other mechanical and chemical insults leads to an increased production of free radicals [12–14]. In addition, our bodies constantly produce free radicals as a by-product of normal aerobic respiration, metabolism and inflammation [14]. Excessive amounts of free radicals from both extrinsic and intrinsic sources can damage the skin stem cells, resulting in a decrease in collagen production by fibroblasts [14] and ultimately leading to the characteristic hallmarks of skin ageing: loss of elasticity and wrinkle formation in sensitive areas such as the face and neck [12, 15]. Other common signs of skin ageing are the appearance of dark spots, loss of hydration due to a decreased production of hyaluronic acid and thinning of the blood vessel walls [12, 16, 17].

Nevertheless, while the deleterious effects of excessive free radicals in the human body have been well established, a more nuanced picture has emerged recently, in which reactive oxygen species produced at lower concentrations can act adaptively by promoting normal/natural apoptosis (cell death) or expression of genes coding for antioxidant enzymes; this phenomenon is known as hormesis [9, 17].

To maintain skin homeostasis, cells lost during turnover are replenished by stem cells generated in the basal layer [18]. However,

epidermal cell renewal slows as we age, as can be demonstrated by monitoring the disappearance of a marker dye from the stratum corneum [19]. Thus, the renewal cycle is around 20 days in young adults, while in older adults it is more than 30 days [2, 20]. Over time, the body's ability to replenish collagen decreases by about 1.0%–1.5% per year, resulting in the appearance of fine lines and deeper wrinkles [5].

Targeting Skin Ageing from within: The Role of Micronutrients

Skin ageing can be targeted proactively, through the use of nutrients that offer protection from oxidative stress, or reactively, by delivering nutrients to the skin or replenishing it with key structural molecules [5, 9, 11]. To protect itself from the attack of free radicals, the skin employs several antioxidant agents, including superoxide dismutase, catalase, glutathione peroxidase, vitamin C and vitamin E [13, 14, 21]. Antioxidants can be broadly grouped into two networks: an endogenous one relying on enzymes, and a non-enzymatic one utilising micronutrients obtained through diet and/or supplementation [9].

Delivering nutrients to the skin can be achieved either through topical treatments or use of oral supplements [5]. In contrast to topical cosmetics, which penetrate the epidermis in small amounts and do not reach the stem cells in the basal layer [22–24], dietary supplements are absorbed systemically and are thus able to target the deepest skin layers from the inside [12]. Combining these two modalities of delivery can provide a holistic skincare regimen, acting from both the inside and the outside [12].

Dietary micronutrients that can potentially benefit the skin include fat-soluble vitamins A (specifically its precursor, beta-carotene), D and E; water-soluble vitamins B and C; and trace minerals such as zinc and selenium. Vitamins A and D are involved in cell differentiation and specialisation, while vitamin E acts as a chain-breaking antioxidant in membranes and lipoproteins [25]. Vitamin E and selenium have synergistic effects in the body, with the first acting as a direct scavenger of lipid radicals and the second as a cofactor for the glutathione peroxidase enzyme [9]. Vitamins E and C also act synergistically, with ascorbic acid acting on the tocopheroxyl radical resulting from cellular oxidative processes to regenerate α-tocopherol [21]. In addition to its role as a potent antioxidant, vitamin C is a cofactor for enzymes involved in maintaining the stable triple-helix conformation of collagen fibres [5] and was shown in pre-clinical studies to increase collagen synthesis and reduce oxidative stress parameters following musculoskeletal injuries [26]. The B-group of vitamins play key roles in energy metabolism, immunity and DNA repair [27].

Clinical research into supplementation as a method to improve skin health and delay the visible signs of ageing has gained momentum, with promising results emerging from clinical trials [5, 21, 28]. Amongst these, a double-blind study of 50 female participants aged between 35 and 65 years with mild-to-moderate photoaging assessed an oral supplement containing antioxidant vitamins (A, C and E), trace minerals (selenium, magnesium and zinc), a phytoestrogen source (flaxseed oil) and coenzyme Q10 for its effects on skin appearance and health [28]. Both investigators and study participants noted significant improvements in wrinkle breadth, skin firmness, hydration and dullness, and subjects' quality of life measures from month 2 of administration onwards [28].

In another placebo-controlled study, in 39 subjects with healthy skin, supplementation with carotenoids, vitamin E and selenium significantly increased skin density and thickness, and improved skin roughness and scaling [29]. The combination of vitamin E and selenium at nutritional doses was shown to provide photoprotective effects, enhance the skin's antioxidative status and improve skin radiance, moisture and elasticity when taken together with carotenoids and polyphenols [25, 29–31]. Vitamins E and C administered together in human volunteers proved effective when used prophylactically against UVB-induced epidermal damage [32].

Riboflavin (vitamin B2) deficiency may increase the risk of inflammatory skin disorders and degenerative diseases in ageing subjects [27, 33], while niacinamide (a vitamin B3 amide analogue) showed clinical benefits in a series of conditions from skin ageing and

hyperpigmentation to acne vulgaris, actinic keratosis and psoriasis, as well as in skin cancer prevention [34–36].

Replenishing Key Structural Molecules in the Skin

The science behind oral collagen supplementation as an effective approach to skin rejuvenation has only recently been elucidated [5]. On ingestion, collagen peptides pass through the gastrointestinal tract, where they are enzymatically broken down into smaller molecular weight fragments such as di- and tripeptides, as well as free amino acids. These are transported across the intestinal wall and readily absorbed into the bloodstream, travelling through the body to reach the sites of dermal fibroblasts [37]. Amino acids such as glycine, proline and hydroxyproline provide building blocks for further synthesis of collagen fibres. At the same time, collagen oligopeptides activate fibroblast receptors and stimulate the production of new collagen and elastin proteins through the inhibition of metalloproteinases 1 and 2, which are involved in collagen degradation [38].

A series of studies in animal models [39, 40] have demonstrated the feasibility of collagen peptide administration in improving collagen density and diminishing the UV-induced collagen reduction in the dermis. In humans, ingestion of bioactive collagen peptides was clinically shown to provide significant improvements in wrinkle appearance and skin elasticity [41–45]. A review and meta-analysis of 19 randomised, placebo-controlled trials with a total of 1125 participants (95% women) found that ingestion of hydrolysed collagen for 90 days was effective in reducing wrinkles and improving skin elasticity and hydration [46].

Hyaluronic acid is another target molecule in skin ageing. Owing to its high water retention and viscosity, hyaluronic acid helps lock in moisture, provides lubrication and cushioning, and maintains fluidity and flexibility in the epidermis, articular cartilage and synovial fluid. Older skin shows a marked disappearance of epidermal hyaluronic acid, although the reasons for this change in homeostasis is not fully understood [16]. In humans, ingestion of hyaluronic acid was clinically shown to boost skin hydration [41]. A recent study with 129 participants confirmed these findings in different age groups and skin types, demonstrating significant improvements in skin hydration after 2–8 weeks of oral supplementation [47].

The cosmetic effects of the two key anti-ageing targets, collagen and hyaluronic acid, were confirmed in a systematic review and meta-analysis of 66 randomised controlled trials (RCTs) published in 2020–2021, examining the effects of dietary supplements on skin moisturisation [48]. In this analysis, which involved a total of 4090 subjects, supplementation with collagen and ceramides resulted in a statistically significant increase in skin hydration and a decrease in trans-epidermal water loss (TEWL) compared with placebo. A statistically significant improvement in skin hydration was also observed with hyaluronic acid supplementation, though its effect on TEWL could not be assessed due to insufficient data [48].

Despite the accumulating evidence, more data and in-depth analyses of the cosmetic effects of collagen and multi-micronutrient supplementation are needed to confirm the benefits of this approach in the clinical setting and its applicability to everyday skincare routines.

MATERIALS AND METHODS

Objective

The objective of this clinical study was to investigate the beneficial effect of a food supplement with collagen peptides and micronutrients on the quality of the skin and general wellbeing of healthy women after an intake period of 12 weeks.

Participants

Healthy female participants ($n = 135$) between 45 and 65 years of age with Fitzpatrick skin

types I–VI, minimum global wrinkles score of 4, presenting with uneven skin tone, dull skin and lax/loose skin were enrolled in the study after having given written informed consent. Subjects were excluded in case of pregnancy or use of food supplements, anti-wrinkle products or hormonal treatments in the month prior to the active phase of the study and prior surgery under general anaesthesia 1 month before the washout phase. The full inclusion and exclusion criteria can be found in the online supplement.

Study Design

The study was conducted as an open-label clinical trial at two sites (Chelmsford and Manchester, United Kingdom). Subjects were pre-screened through telephone interviews, and if eligible, were enrolled in a 28-day washout phase. This washout period was chosen on the basis of the available reports from skin studies with topical or oral supplementation [49, 50] and is in alignment with the 20–30 day transit time for stratum corneum, the layer acted on by most topical skincare products [51].

Subjects attended all visits bare-faced, without make-up (including mascara). A facial cleanser (Dove soap) and facial moisturiser (Olay Beauty Fluid Face & Body Moisturising Fluid) were provided at visit 1 (washout phase). Subjects agreed to replace their facial cleansing/moisturising products with those provided and not to introduce any new skincare products, cosmetics or supplements for the duration of the study. Further, subjects agreed not to use sunbeds or UV lamps, and avoided prolonged sun exposure during the study period. The washout phase was followed by 3 months of treatment (active phase). Visit 2 was the baseline visit; follow-up visits (visits 3, 4 and 5) were scheduled at 1, 2 and 3 months of treatment, respectively.

Participants were required to return the empty packaging, including blisters, at each visit. Compliance checks were carried out via weekly diaries asking participants to confirm they had used the product as instructed and record any missed dose. Participants were also asked a number of questions on compliance as part of the subjective questionnaire.

Endpoints

The primary endpoint of the study was the change from baseline in global wrinkles at month 3 by expert grader analysis. Secondary endpoints included: change from baseline in global wrinkles at months 1 and 2; changes from baseline in visual and tactile grading of global fine lines, skin elasticity, skin tone evenness, radiance, overall skin quality/healthy appearance, firmness, texture and Glogau score; elasticity (R2) by Cutometer; subjective assessment of product questions; and total antioxidant status (TAS).

The mixture of objective and subjective measures (Cutometer, expert grader analysis, participant opinion), as well as the large sample size, bring robustness to the study design, even though the lack of control group is a limitation.

Test Product

The product used in this study was Richelet Skin Renewal (P&G Health), a supplement with collagen peptides, vitamins and minerals formulated to assist the maintenance of healthy skin and to reduce the visible signs of skin ageing. The test product was distributed to the study participants as three 1500 mg tablets per day. Tablets were taken together orally with a glass of water, once a day. The daily dosage of the study product contained 2.5 g collagen peptides (Verisol®) and 120 mg hyaluronic acid. Other ingredients include vitamins A, C, D and E, and minerals zinc and selenium (Table S1 of the online supplement) [52].

Assessments and Procedures

At each visit during the active phase, subjects had high-resolution clinical photographs (frontal view, 90° right, 90° left) taken by an expert grader. They were invited to acclimatise in a controlled environment (temperature of 20–24 °C and relative humidity 35–50%) for 30 min. The expert grader then visually assessed each subject on a scale of 0–9 (0.5 increments)

for attributes such as: global wrinkles, global fine lines, skin elasticity, skin tone evenness, skin radiance/luminosity, overall skin quality/healthy appearance and Glogau scale (included in the online supplement) [53]. The grader also completed tactile assessments for skin firmness (overall and around eyes) and skin texture/smoothness.

Subjects then underwent bio-instrumentation assessments for elasticity via R2-Cutometer measurement below the subject's right eye (middle malar region). Skin elasticity was measured using a Cutometer® MPA 580 (Courage and Khazaka, Germany). Extension of the skin is measured by creating negative pressure in the device, drawing the skin into the aperture of the probe and using a non-contact optical system to determine the penetration depth inside the probe. This is followed by a 3-s relaxation time for measuring elasticity (R2); an average of more than three repetitions is reported, providing a reliable and accurate measure of the skin's viscoelastic properties.

Subjects then completed a self-perception questionnaire (SPQ). The questionnaire was composed of 26 questions with many multi-part questions, covering usage compliance, skin benefits and wellbeing, a combination of which were asked at each study visit.

Finally, subjects had their blood taken for TAS analysis; antioxidant levels in serum were measured spectrophotometrically (at 600 nm wavelength) using the commercially available assay by Randox Laboratories Ltd (Crumlin, UK). On completion of these assessments, subjects were given enough supplements to last until their next scheduled visit, as well as an information sheet along with a diary to use at home.

Any adverse events were documented during the study intervention.

Data Analysis and Statistics

A sample size of 150 was chosen to provide 90% power to detect a difference in means of 0.33 in the expert grader scale, assuming a standard deviation of differences of 1.18, using a paired *t*-test with a 5% two-sided significance level and 10% inflation for dropouts. Subject demographics and baseline characteristics were summarised by descriptive statistics.

The primary efficacy endpoint and secondary endpoints (expert grader, Glogau score and change from baseline in SPQ) violated assumptions of normality, so non-parametric testing was done using the Wilcoxon signed-rank test. For secondary parameters such as elasticity by Cutometer and TAS, which satisfied normality assumptions, the paired *t*-test was used for testing. The after-use agreement questions were analysed by binomial testing if agreement (agree/strongly agree or top 2 box/categories) differed from a probability of 0.5.

Compliance with Ethics Guidelines

This study was granted approval by the East Anglia Ethical Committee on 23 February 2023 and was conducted in accordance with the guidelines set forth by the International Conference of Harmonization Guidelines for Good Clinical Practice, and the Declaration of Helsinki regarding the treatment of human subjects in a study. Prior to the conduct of any study-specific procedures, subjects were provided an explanation of the nature of the study, including the purpose, procedures, expected duration and potential risk. Prospective subjects were informed of their right to withdraw from the study at any time without being obliged to give a reason. If consent was obtained, the subject signed and dated the informed consent form.

RESULTS

Subjects and Dropouts

The enrolment population was initially planned at 160, for a minimum of 150 subjects receiving the test product. A total of 168 subjects were recruited and began the pre-study washout phase, with 33 washout drops (not meeting inclusion criteria or not continued per investigator request), resulting in 135 subjects enrolled between the two sites and receiving the test product. Ultimately, 116 of the 135

Table 1 Participant disposition (intent-to-treat population)

Disposition	***n*** **(%)**
Received study product (*N* = 135)	
Completed	116 (85.9%)
Discontinued	19 (14.1%)
Reason for discontinuation	
Voluntary withdrawal	11 (8.1%)
Lost to follow-up	7 (5.2%)
Adverse event	1 (0.7%)

Table 2 Demographic and baseline characteristics of study subjects

Characteristic (***N*** **= 135**)	
Age, years	
Mean (SD)	54.5 (6.18)
25th percentile	49
75th percentile	60
Median	54
Min–max	42–65
Fitzpatrick scale	***n*** **(%)**
I	19 (14.1%)
II	99 (73.3%)
III	5 (3.7%)
IV	5 (3.7%)
V	6 (4.4%)
VI	1 (0.7%)

SD standard deviation

participants completed all aspects of the study, while 19 discontinued the study during the treatment phase (Table 1). Subjects had a mean age of 54.5 years at baseline; 73.3% had Fitzpatrick skin type II (Table 2).

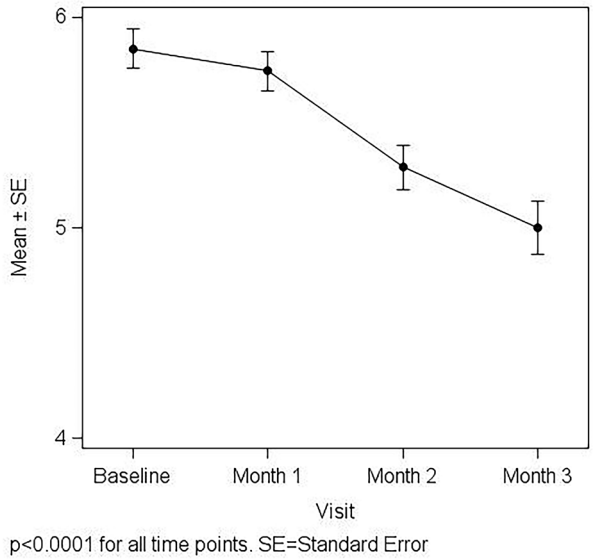

Fig. 1 Mean expert grader global wrinkles by visit

Global Wrinkles by Expert Grading

The primary efficacy outcome of mean global wrinkles score by expert grader analysis demonstrated a statistically significant decrease from 5.9 at baseline to 5.0 at month 3 ($p<0.0001$; Fig. 1), with 83.6% of subjects showing an improvement (Table 3). At the month 3 visit, the median reduction in global wrinkles was 16.7%, with 15% of the population experiencing at least a 33.3% reduction and maximum 50.0% reduction observed. Therefore, the study met its primary endpoint as defined in the study protocol. An example photo showing wrinkle reduction with average 3 month treatment response in shown in Fig. 2.

The secondary efficacy endpoint of change from baseline to months 1 and 2 in global wrinkles was also met, with the analysis of visual and tactile grading of global wrinkles indicating statistically significant decreases from baseline to all subsequent timepoints ($p < 0.0001$; Fig. 1). In addition, the global fine lines, skin elasticity, skin firmness, skin texture/smoothness, skin tone evenness, skin radiance/luminosity and overall skin quality secondary endpoints all showed statistically significant improvements at each timepoint

when compared to baseline ($p < 0.0001$; Table 4). Notably, 100% of subjects had visual/tactile improvements in skin texture/smoothness, skin tone evenness, skin radiance/luminosity and overall skin quality at month 3 (Table 4). All other skin attributes had at least 94.0% of subjects improving at the month 3 visit (Table 4).

Skin Elasticity

Analysis of R2 values from Cutometer measurements revealed no changes from baseline to month 1, while 55.7% of subjects showed improvements at month 2, corresponding to an increase in the mean R2 score from 0.67 to 0.69 (Table 5). The improvement observed at month 3 (R2 score 0.74) was statistically significant ($p < 0.0001$), with evidence of increased skin elasticity after 3 months of supplementation in 63.8% of subjects (Table 5). At the month 3 visit, the median improvement in skin elasticity measured by Cutometer was 12.5%, with 15% of the population experiencing at least 47.1% improvement.

Glogau Scale

Analysis of Glogau scores showed statistically significant improvements at month 2 ($p = 0.0039$) and month 3 ($p = 0.0002$); no change was observed at month 1.

Antioxidant Status

Within-treatment TAS analysis showed no significant difference at any timepoint compared with baseline. Scores at baseline were 1.729 and increased slightly to 1.746 at month 3.

Subjective Questionnaires

SPQ data were collected showing the percentage of responses to each question/statement. The questions were extensive, so a red herring question was asked at the final visit to gauge accuracy and compliance. Analyses were performed on the subset of 71 of 116 subjects who selected the correct red herring response. In examination of the red herring correct subset, nearly 3/4 (74%) of self-perception questions reached statistical significance.

The SPQs conveyed favourable subjective responses regarding the use of the study product (Tables S2–S6 of the online supplement). The self-rating of overall skin appearance improved significantly at all timepoints versus baseline ($p < 0.0001$ for all), with 83.1% of participants rating it favourably at month 3. The self-rating of appearance of fine lines and wrinkles improved significantly at months 1 ($p = 0.005$), 2 and 3 ($p < 0.0001$ for both) versus baseline, with 78.9% of participants rating their appearance favourably at month 3. The majority (67.6%) of respondents agreed that wrinkle appearance around the eyes was reduced after 3 months of supplementation. The self-rated feeling of skin firmness also improved significantly at all timepoints ($p < 0.0001$), and 74.6% of participants agreed their skin was left feeling firmer after 3 months. The self-rating of skin texture, radiance, skin tone and brightness all improved significantly at all timepoints. At month 3, 85.9% of participants rated their skin

Table 3 Change from baseline in global wrinkles grading by visit (all subjects)

Visit	*n*	Mean (SD)	Median	Min–max	*n* (%) Improved	*p*-value*
Global wrinkles						
Month 1	127	−0.2 (0.30)	0.0	−1.0 to 0.0	30 (23.6%)	< 0.0001
Month 2	122	−0.6 (0.53)	−0.5	−2.0 to 0.0	86 (70.5%)	< 0.0001
Month 3	116	−1.0 (0.71)	−1.0	−3.0 to 0.0	97 (83.6%)	< 0.0001

*Wilcoxon signed-rank *p*-value, *SD* standard deviation

Fig. 2 Visible reduction of eye, forehead and glabella wrinkles after 3 months' intake of Richelet Skin Renewal supplement. Example pictures of participant with average treatment response

texture favourably, 76.1% agreed their skin was smoother, 80.3% thought the visible texture of their skin was better and 80.3% rated skin radiance/glow favourably.

Additionally, 73.2% of participants agreed the product worked with their skin tone, with 85.9% rating skin tone and 83.1% rating brightness favourably at month 3. The product also significantly improved the feeling of skin moisturisation at all timepoints ($p < 0.0001$), with 83.1% of participants rating it favourably. After 3 months of supplementation, 78.9% of participants agreed the product left their skin feeling moisturised, 77.5% agreed the product was effective and 69.0% agreed it was a great addition to their skincare routine.

Safety Outcomes

Overall, the study product was well tolerated. Only one adverse event (red rash on the face) was reported during the study, which was classified as mild. The study product was discontinued, and the adverse event resolved without the use of concomitant medication. The subject chose to withdraw from the study.

DISCUSSION

Clinical Context and Implications for Practice

Daily skincare routines and regimens can have a significant long-term impact on the overall quality of a person's skin [54]. Owing to an ageing population with a high awareness of the importance of skin health and a desire for sustainable skincare solutions, concepts such as 'beauty-from-within' have become important components of the modern, holistic approach to skincare [55]. Dermatologists and consumers alike are increasingly aware of the deleterious effects of oxidative stress on our bodies and the decrease in collagen and hyaluronic acid production over time, all of which result in wrinkle formation, loss of elasticity and other visible signs of skin ageing [46]. The use of injectable treatments, particularly dermal fillers such as hyaluronic acid, has increased over the years, and while still low overall, the number of side effects has also risen [56]. Reported adverse reactions to dermal fillers include pain, erythema, infection and hypersensitivity [57]. Despite hyaluronic acid fillers being structurally

Table 4 Change from baseline in skin visual and tactile grading by visit (all subjects)

Visit	*n*	Mean (SD)	Median	Min–max	*n* (%) Improved	*p*-Value*
Global fine lines						
Month 1	127	−0.3 (0.36)	−0.5	−1.0 to 0.0	64 (50.4%)	< 0.0001
Month 2	122	−0.8 (0.56)	−1.0	−2.5 to 0.0	101 (82.8%)	< 0.0001
Month 3	116	−1.2 (0.71)	−1.0	−3.0 to 0.0	109 (94.0%)	< 0.0001
Skin elasticity						
Month 1	127	−0.5 (0.34)	−0.5	−1.5 to 0.0	95 (74.8%)	< 0.0001
Month 2	122	−1.2 (0.55)	−1.0	−2.5 to 0.0	118 (96.7%)	< 0.0001
Month 3	116	−1.6 (0.72)	−1.5	−3.5 to 0.0	115 (99.1%)	< 0.0001
Skin firmness						
Month 1	127	−0.3 (0.35)	−0.5	−1.5 to 0.0	71 (55.9%)	< 0.0001
Month 2	122	−1.0 (0.67)	−1.0	−3.5 to 0.0	110 (90.2%)	< 0.0001
Month 3	116	−1.5 (0.76)	−1.5	−3.5 to 0.0	115 (99.1%)	< 0.0001
Skin texture/smoothness						
Month 1	127	−0.8 (0.46)	−1.0	−2.0 to 0.0	111 (87.4%)	< 0.0001
Month 2	122	−2.3 (0.61)	−2.0	−4.0 to −1.0	122 (100%)	< 0.0001
Month 3	116	−3.2 (0.82)	−3.0	−5.0 to −1.5	116 (100%)	< 0.0001
Skin tone evenness						
Month 1	127	−0.6 (0.50)	−0.5	−2.5 to 0.0	86 (67.7%)	< 0.0001
Month 2	122	−1.7 (0.78)	−2.0	−4.0 to 0.0	119 (97.5%)	< 0.0001
Month 3	116	−2.5 (1.04)	−2.0	−6.0 to −1.0	116 (100%)	< 0.0001
Skin radiance/luminosity						
Month 1	127	−0.7 (0.43)	−0.5	−2.0 to 0.0	105 (82.7%)	< 0.0001
Month 2	122	−2.2 (0.63)	−2.0	−4.0 to −1.0	122 (100%)	< 0.0001
Month 3	116	−3.1 (0.83)	−3.0	−6.0 to −1.0	116 (100%)	< 0.0001
Overall skin quality						
Month 1	127	−0.4 (0.37)	−0.5	−1.5 to 0.0	85 (66.9%)	< 0.0001
Month 2	122	−1.4 (0.60)	−1.5	−3.0 to 0.0	120 (98.4%)	< 0.0001
Month 3	116	−2.3 (0.97)	−2.0	−5.0 to −0.5	116 (100%)	< 0.0001

*Wilcoxon signed-rank *p*-value; *SD* standard deviation

Table 5 Analysis of elasticity (R2-Cutometer) by visit (all subjects)

Visit	*n*	Mean (SD)	Median	Min–max	*n* (%) Improved	*p*-Value*
Elasticity						
Baseline	135	0.67 (0.135)	0.67	0.32–0.96		
Month 1	127	0.67 (0.133)	0.69	0.28–0.96	60 (47.2%)	0.7904
Month 2	122	0.69 (0.128)	0.70	0.08–0.95	68 (55.7%)	0.1743
Month 3	116	0.74 (0.102)	0.76	0.33–0.95	74 (63.8%)	< 0.0001

*Paired *t*-test *p*-value, *SD* standard deviation

similar to endogenous hyaluronic acid and thus having low immunogenicity, a number of cases of delayed inflammatory reactions were reported, some of which were related to the coronavirus disease 2019 (COVID-19) spike protein [57–60]. Use of supplementation as a strategy to support skin appearance could be an alternative strategy to dermal injections, as supplements are generally well tolerated. With the increased interest in supplementation as a strategy to improve skin health and appearance, clinical research into collagen supplements has been steadily growing [46, 48]. However, some clinical trials have been inconclusive, highlighting the need for further investigations to fully elucidate the cutaneous effects of oral collagen supplementation [45, 61].

The study product supports cell renewal from within the body and reduces the signs of skin ageing through the use of collagen peptides, hyaluronic acid and micronutrients (vitamins and minerals). The key antioxidants in this supplement are of natural origin, with soybean used as a natural source of vitamin E and inactivated *Saccharomyces cerevisiae* as an organic source of selenium. Vitamin A, vitamin D and zinc, as well as some other vitamins and minerals, have roles in the cell cycle, for example, in cell division and differentiation [62, 63]. Vitamin C is a strong antioxidant and helps maintain the stability of collagen fibres [5]. The action of vitamin E is boosted by co-administration of selenium [25] and Vitamin C [64]. The use of vitamins and minerals alongside structural molecules can enhance the overall effectiveness of a supplement, as micronutrients act in synergy in the body [9, 21]. Further, when used in combination in the clinical setting, multi-micronutrient formulations helped improve skin appearance, provide photoprotection and mitigate the signs of ageing [25, 29–32]. Compared with other supplements discussed in the literature [46–48], the study product has a comprehensive formula, acting on the skin via numerous mechanisms of action, which support renewal of the cells, protect them from free radicals and replenish key structural molecules such as collagen and hyaluronic acid. This may explain the level of improvement in overall skin appearance observed in this study by comparison with other reports [46, 48].

The present study of the Skin Renewal food supplement showed statistically significant improvements reported on the efficacy measures of change from baseline in global wrinkles by expert grading at month 3, as well as at the intermediate timepoints (i.e. months 1 and 2). The majority of subjects (83.6%) showed an improvement in the global wrinkles score after 3 months. Furthermore, all subjects demonstrated significant improvements in attributes such as skin texture, skin tone evenness, skin radiance and overall skin quality at the month 3 visit. Use of the supplement for 3 months resulted in statistically significant increases in skin elasticity by R2-Cutometer in 63.8% of the subjects.

The objective and subjective assessments were consistent, and the effects were noticed by both the expert grader and the participants themselves. Subjective data from SPQs corroborated the objective findings, with statistically significant improvements in the self-rating of overall skin appearance, fine lines and wrinkles, skin texture, radiance and glow

recorded at all timepoints. These benefits were meaningful for the participants, as confirmed by a majority of respondents rating the product favourably at 12 weeks on all these aspects.

The TAS secondary endpoint did not show significant changes. The baseline values were in the high end of the normal expected range provided by the assay supplier (range 1.30–1.77 mmol/l), but it was not possible to ascertain whether the lack of response after treatment was due to assay sensitivity, choice of antioxidant test or other factors. However, it should be noted that TAS is not directly related to beauty outcomes.

Placed in context, the improvements in global wrinkles and skin elasticity observed here strengthen the evidence base for the efficacy of oral collagen supplements. In a double-blind, placebo-controlled clinical trial of 114 women aged 45–65 years, 4 weeks of treatment with bioactive collagen peptides (Verisol®) 2.5 g once daily resulted in a statistically significant reduction in eye wrinkle volume versus placebo; the effect was more pronounced after 8 weeks of intake, with a maximum reduction in eye wrinkle volume of 49.9% [43]. A second double-blind, placebo-controlled trial investigated the effect of two doses of Verisol®, 2.5 g and 5 g, versus placebo in 69 women aged 35–55 years [44]. Both regimens produced a statistically significant improvement in skin elasticity compared with placebo at 4 and 8 weeks, with no significant differences in outcomes between the two doses.

The other anti-ageing ingredient in the test product, hyaluronic acid, was clinically proven in previous studies to boost skin moisture and improve facial ageing symptoms [41, 42]. Placebo-controlled trials in Japanese subjects found that consumption of hyaluronic acid at the clinically relevant dose of 120 mg, administered orally for 6–12 weeks, significantly increased skin moisture content and improved wrinkle appearance both by dermatologist assessment and by subjective evaluations [41, 42]. In a further placebo-controlled trial in 129 participants of varying age and skin type (oily, normal and dry), consumption of 100 mg hyaluronic acid for 12 weeks significantly improved skin hydration [47].

Of note, one tablet of the test product contains 2.5 g collagen and 120 mg hyaluronic acid. Therefore, the dosing of these key ingredients in the present study (three tablets daily) as well as the duration of the active treatment phase (3 months) are comparable with previous reports [41–44]. Additionally, the molecular weight of hyaluronic acid influences its penetration into the skin and its biological activity [65]. High molecular weight hyaluronic acid is unable to penetrate the skin; therefore, oral supplementation could have delivery advantages and help drive skin benefits.

Taken together, the results summarised here build a coherent and convincing argument for collagen and multi-micronutrient supplementation as an effective approach to improving skincare outcomes in a clinical setting, with the potential to benefit everyday practice and beauty regimens on a broader scale.

Study Strengths and Limitations

The present study provides a comprehensive evaluation of the test product with a wide range of endpoints reported and analysed. The sample size is comparable to or greater than that used in previous studies [41–44] and the measurements include both objective and subjective assessments of skin appearance and wellbeing.

The main limitation of this study is its open-label design, which limits the quality of the evidence produced by comparison with RCTs. Herein, open-label use of the oral supplement alongside a topical skincare routine was deemed representative of real-world situations. Further, despite the lack of a placebo, the benefit of the test product over time was demonstrated by subjective measures (participant opinion), expert graders and objective assessment of skin elasticity (Cutometer).

Other than a prohibition on the use of all other food supplements (including vitamin supplements) from the beginning of the washout phase and throughout the study, for practical reasons there was no control or monitoring of diet in this study. Therefore, changes in dietary

sources of collagen could have accounted for some of the effects observed in the study, though it is unlikely they would have accounted for the significant improvements from baseline across a large group of participants. Currently, the reported ability to assess nutrient intake is poor, as only some of the nutrients have good biochemical status indicators [66] and subjects are prone to over- or under-reporting their food intake [67].

CONCLUSIONS

In this clinical study, the Richelet Skin Renewal supplement with collagen peptides, hyaluronic acid and micronutrients provided statistically significant, noticeable improvements in global wrinkles, skin elasticity and a range of other skin attributes with 3 months of use in healthy women. Skin benefits were demonstrated consistently through both objective and subjective assessments. This study adds to the body of evidence highlighting the potential skin benefits of oral supplementation with collagen peptides, hyaluronic acid and micronutrients as part of a holistic approach to skincare.

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Data Availability. The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of Interest. Rachael Gibson, Lieselotte Krug, David L Ramsey, Azadeh Safaei and Sue Aspley are employees of P&G Health.

Ethical Approval. This study was granted approval by the East Anglia Ethical Committee on 23 February 2023 and was conducted in accordance with the guidelines set forth by the International Conference of Harmonization Guidelines for Good Clinical Practice, and the Declaration of Helsinki regarding the treatment of human subjects in a study. Prior to the conduct of any study-specific procedures, subjects were provided an explanation of the nature of the study, including the purpose, procedures, expected duration, and potential risk. Prospective subjects were informed of their right to withdraw from the study at any time without being obliged to give a reason. If consent was obtained, the subject signed and dated the informed consent form.

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